



Universidade de Brasília
Departamento de Estatística

Redução de Dimensionalidade: Análise Fatorial

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Resolução da Lista 6 da disciplina Análise
Multivariada.

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1 Teoria

$$h_i^2$$

é comunalidade, calculada pela soma dos quadrados dos valores de das cargas fatoriais (ℓ_{ij})

$$\psi_i$$

representa o erro ou a parte daquela variável que é única e não tem nada a ver com os outros fatores, é o que a comunalidade não explica, variância específica.

$$Var(X_i) = h_i^2 + \psi_i$$

2 Questões

2.1 4.1 Artes e Barroso

Clientes de uma rede de lojas do setor de vetuário avaliaram a qualidade dos pontos de venda (dados fictícios). Foram atribuídas notas (de 0 a 10) aos seguintes atributos (quanto maior a nota, maior é a percepção de que o ponto de venda possui atributo)

- X1: Cordialidade dos vendedores;
- X2: Limpeza da Loja;
- X3: Falta de agilidade no atendimento;
- X4: Demora na concessão de crédito;
- X5: Vendedores bem informados sobre os produtos da loja.

Para entender o padrão de respostas observado na amostra, aplicou-se uma análise fatorial à matriz de correlação dessas variáveis. A tabela 4.18 apresenta as cargas fatoriais obtidas após uma rotação varimax.

Tabela 1: Tabela 4.18 do livro Artes e Barroso

	F1	F2	h^2	ψ
X1	-0.790	0.4300	0.809	0.191
X2	-0.070	0.9600	0.9265	0.0735
X3	0.900	-0.1800	0.8424	0.1576
X4	0.920	0.2200	0.8948	0.1052
X5	-0.640	0.4000	0.5696	0.4304
SS	2.695	1.3473		

a) Determine e interprete as comunalidades e as porcentagens de explicação dos fatores. Critique a escolha do número de fatores

F1 é definido positivamente por X3 e X4, e negativamente por X1 e X5. Portanto, podemos apontar F1 como Ineficiência do atendimento;

F2 é definido positivamente, com uma carga altíssima, por X2, Logo, podemos

apontar F2 como Ambiente da Loja.

A comunalidade de X2 é a maior, indicando que a percepção de limpeza é quase totalmente explicada pelos constructos do modelo, sendo um indicador confiável para o “Ambiente da Loja”. Já a de X5, que é a menor, indica que os constructos explicam apenas 56.96% da variável, o restante é dado pela variância específica, não capturada pelos constructos.

A porcentagem de explicação é de 0.539 para F1 e 0.269 para F2, totalizando 0.808. o que é satisfatório, pois retém a maior parte da informação original com apenas dois índices. Logo, $m = 2$ (2 fatores) é suficiente, uma vez que se busca 70-80% em análises exploratórias.

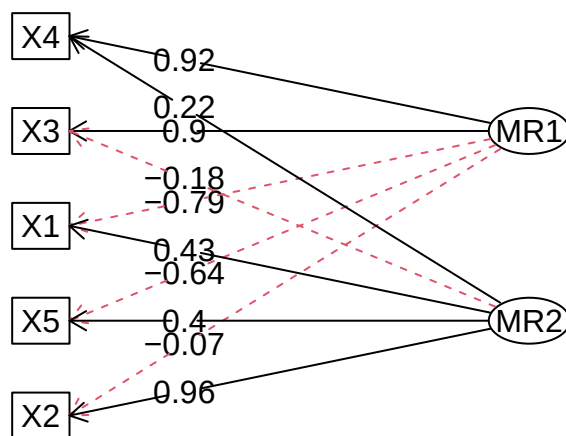
b) Interprete os fatores rotacionados

O problema já diz que a tabela 4.18 é proveniente da rotação de varimax. Desta maneira, é possível observar com mais clareza quais variáveis de fato dominam um constructo. Por exemplo: X2 (limpeza) possui uma carga de 0.96 no Fator 2, e quase zero no Fator 1, esse comportamento se deve a rotação, esse é um fator rotacionado, onde toda a explicação da limpeza foi “puxada” para uma coluna.

c) Construa um diagrama de caminho que descreva a análise realizada. Descreva as respectivas equações.

Figura 1: Diagrama Fatorial 4.1

Diagrama Fatorial



As equações são:

$$X_1 - \mu_1 = -0.790F_1 + 0.430F_2 + \varepsilon_1$$

$$X_2 - \mu_2 = -0.070F_1 + 0.926F_2 + \varepsilon_2$$

$$X_3 - \mu_3 = 0.900F_1 + 0.842F_2 + \varepsilon_3$$

$$X_4 - \mu_4 = 0.920F_1 + 0.894F_2 + \varepsilon_4$$

$$X_5 - \mu_5 = -0.640F_1 + 0.569F_2 + \varepsilon_5$$

d) Um analista reservou os dois fatores e determinou seus escores fatori-

ais. O diagrama de dispersão apresentado na Figura 4.7 descreve o comportamento desses escores. Nesse diagrama foi destacado um ponto (vide a flecha) referente a um determinado consumidor. Como você interpretaria a percepção desse consumidor em relação aos demais? O analista conclui que se tratava de um valor aberrante multidimensional. Essa conclusão faz sentido?

Dada posição desse consumidor, que é próximo de 0 no escore fatorial do fator 1 e perto de -4 no escore fatorial do fator 2, é possível dizer que a percepção dele acerca da ineficiência do atendimento é mediana, ou que a agilidade e cordialidade estão “na média”. Porém, por outro lado, esse indivíduo tem uma percepção extremamente negativa do ambiente da loja (4 desvios padrões abaixo da média), o que o torna inconsistente em relação às outras observações.

2.2 4.3 Artes e Barroso

Retome o enunciado do Exercício 2.5. Aplique uma análise fatorial pelo método das componentes principais utilizando a matriz de correlações policóricas. O conjunto de fatores deve explicar pelo menos 70% da variabilidade total dos dados. Faça uma análise completa dos resultados.

A tabela 2.10 traz os dados da aba Medias de Pizza.xlsx, com informações sobre os valores médios e características físico-químicas avaliadas em amostras de dez marcas de pizzas. A lista das características está na tabela 2.9.

Tabela 2: Tabela 2.10 do livro Artes e Barroso

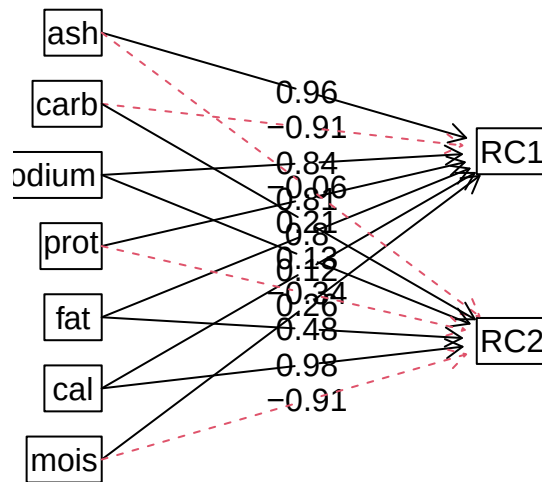
	F1	F2	h^2	ψ
mois	0.26	-0.91	0.9019	0.0981
prot	0.81	-0.34	0.7786	0.2214
fat	0.80	0.48	0.8689	0.1311
ash	0.96	-0.06	0.9235	0.0765
sodium	0.84	0.13	0.7296	0.2704
carb	-0.91	0.21	0.8675	0.1325
cal	0.12	0.98	0.9748	0.0252
SS	3.83	2.21		

O constructo F1 é positivamente dominado por ash, sodium, prot e fat, e negativamente por carb. Portanto, podemos apontar F1 como perfil de ingredientes.

O constructo F2 é positivamente delimitado por cal, e negativamente por mois. Logo, podemos apontar F2 como Calorias X umidade.

A maior comunalidade pertence a Ash, indicando que a percepção acerca do perfil dos ingredientes é quase totalmente explicada pelos constructos do modelo, sendo um indicador confiável. A menor comunalidade pertence ao sódio, ou seja, fatores não capturados pelo modelo a influenciam.

Figura 2: Diagrama Fatorial 4.3

Diagrama Fatorial**2.3 4.4 Artes e Barroso**

Retome o enunciado do Exercício 3.4. Aplique uma análise fatorial pelo método das componentes principais utilizando a matriz de correlações policóricas. O conjunto de fatores deve explicar pelo menos 70% da variabilidade total dos dados. Faça uma análise completa dos resultados.

```
dados <- read.csv("Q4_4.csv") %>% select(-PUF_ID)

dados_ord <- dados %>%
  mutate(across(
    everything(),
    \(x) {
      cortes <- quantile(
        x,
        probs = seq(0, 1, length.out = 11),
        na.rm = TRUE
      )
      cortes <- unique(cortes)

      cut(
        x,
        breaks = cortes,
        include.lowest = TRUE,
        labels = FALSE
      ) - 1
    }
  ))
```

```
matriz_poly <- polychoric(dados_ord, max.cat = 11)$rho
```

```
Item = FK1correct had no variance and was deleted
Item = FK2correct had no variance and was deleted
Item = FK3correct had no variance and was deleted
Item = KHKNOWL6 had no variance and was deleted
Item = KH1correct had no variance and was deleted
Item = KH2correct had no variance and was deleted
Item = KH3correct had no variance and was deleted
Item = KH4correct had no variance and was deleted
Item = KH5correct had no variance and was deleted
Item = KH6correct had no variance and was deleted
Item = KH7correct had no variance and was deleted
Item = KH8correct had no variance and was deleted
Item = KH9correct had no variance and was deleted
Item = IMPUTATION_FLAG had no variance and was deleted
Item = PRODHAVE_1 had no variance and was deleted
Item = PRODHAVE_2 had no variance and was deleted
Item = PRODHAVE_3 had no variance and was deleted
Item = PRODHAVE_4 had no variance and was deleted
Item = PRODHAVE_5 had no variance and was deleted
Item = PRODHAVE_6 had no variance and was deleted
Item = PRODHAVE_7 had no variance and was deleted
Item = PRODHAVE_8 had no variance and was deleted
Item = PRODHAVE_9 had no variance and was deleted
Item = PRODUSE_1 had no variance and was deleted
Item = PRODUSE_2 had no variance and was deleted
Item = PRODUSE_3 had no variance and was deleted
Item = PRODUSE_4 had no variance and was deleted
Item = PRODUSE_5 had no variance and was deleted
Item = PRODUSE_6 had no variance and was deleted
Item = SHOCKS_1 had no variance and was deleted
Item = SHOCKS_2 had no variance and was deleted
Item = SHOCKS_3 had no variance and was deleted
Item = SHOCKS_4 had no variance and was deleted
Item = SHOCKS_5 had no variance and was deleted
Item = SHOCKS_6 had no variance and was deleted
Item = SHOCKS_7 had no variance and was deleted
```

```
Item = SHOCKS_8 had no variance and was deleted
Item = SHOCKS_9 had no variance and was deleted
Item = SHOCKS_10 had no variance and was deleted
Item = SHOCKS_11 had no variance and was deleted
Item = SHOCKS_12 had no variance and was deleted
Item = ON2correct had no variance and was deleted
Item = ON1correct had no variance and was deleted
Item = INTERCONNECTIONS_1 had no variance and was deleted
Item = INTERCONNECTIONS_2 had no variance and was deleted
Item = INTERCONNECTIONS_3 had no variance and was deleted
Item = INTERCONNECTIONS_4 had no variance and was deleted
Item = INTERCONNECTIONS_5 had no variance and was deleted
Item = INTERCONNECTIONS_6 had no variance and was deleted
Item = INTERCONNECTIONS_7 had no variance and was deleted
Item = INTERCONNECTIONS_8 had no variance and was deleted
Item = INTERCONNECTIONS_9 had no variance and was deleted
Item = INTERCONNECTIONS_10 had no variance and was deleted
Item = EMPLOY1_1 had no variance and was deleted
Item = EMPLOY1_2 had no variance and was deleted
Item = EMPLOY1_3 had no variance and was deleted
Item = EMPLOY1_4 had no variance and was deleted
Item = EMPLOY1_5 had no variance and was deleted
Item = EMPLOY1_6 had no variance and was deleted
Item = EMPLOY1_7 had no variance and was deleted
Item = EMPLOY1_8 had no variance and was deleted
Item = EMPLOY1_9 had no variance and was deleted
Item = PPGENDER had no variance and was deleted
Item = PPMSACAT had no variance and was deleted
Item = PPT01 had no variance and was deleted
Item = PPT25 had no variance and was deleted
Item = PPT612 had no variance and was deleted
Item = PPT1317 had no variance and was deleted
Warning in polychoric(dados_ord, max.cat = 11): The items do not have an equal
number of response alternatives, global set to FALSE.
Warning in matpLower(x, nvar, gminx, gmaxx, gminy, gmaxy): 63 cells were
adjusted for 0 values using the correction for continuity. Examine your data
carefully.
Warning in cor.smooth(mat): Matrix was not positive definite, smoothing was
```


done

```
fit <- principal(matriz_poly, nfactors = 30, rotate = "varimax")
print(fit)
```

Principal Components Analysis

```
Call: principal(r = matriz_poly, nfactors = 30, rotate = "varimax")
```

Standardized loadings (pattern matrix) based upon correlation matrix

[illegible]

KHKNOWL4 0.13 0.01	-0.10	-0.02	-0.02	-0.51	-0.04	-0.10	0.03	-0.02	-
KHKNOWL5 0.03	-0.11	-0.01	0.21	-0.43	-0.02	0.02	-0.05	-0.10	0.12 -
KHKNOWL7 0.04	0.13	0.08	-0.08	0.57	0.03	-0.01	-0.06	0.03	0.05 -
KHKNOWL8 0.07	-0.06	-0.07	-0.08	-0.16	-0.05	-0.05	0.01	0.04	-0.07 -
KHKNOWL9 0.15	-0.16	0.04	-0.01	-0.59	-0.03	-0.16	-0.05	-0.02	-0.07 -
ENDSMEET 0.03	-0.82	-0.15	0.04	-0.15	-0.03	-0.10	-0.05	0.05	-0.05 -
HOUSING 0.62	-0.28	-0.10	0.37	-0.19	0.07	-0.21	-0.02	-0.20	-0.02 -
LIVINGARRANGEMENT 0.04	-0.08	-0.01	0.14	-0.01	0.01	-0.07	0.00	0.00	0.01 -
HOUSERANGES	-0.01	-0.04	0.15	-0.04	0.05	0.01	-0.01	-0.02	0.04 0.66
VALUERANGES	0.34	0.12	-0.23	0.18	0.03	0.10	-0.01	0.13	0.15 0.69
MORTGAGE	0.11	0.02	-0.06	0.03	-0.01	0.13	0.00	0.15	0.02 0.89
SAVINGSRANGES	0.55	0.07	-0.13	0.04	0.11	0.02	0.04	-0.10	0.03 0.29
CONSPROTECT1	-0.29	-0.06	0.09	0.05	0.02	0.01	0.03	-0.06	0.09 0.04
CONSPROTECT2	-0.04	0.25	-0.02	-0.08	0.05	0.09	0.08	0.01	0.05 0.06
CONSPROTECT3 0.01	-0.13	0.05	-0.07	-0.19	0.03	0.13	0.04	-0.01	0.10 -
EARNERS	0.03	0.00	-0.01	0.04	0.03	0.11	0.04	-0.05	0.04 0.07
VOLATILITY 0.05 0.05	-0.23	0.01	0.15	-0.12	-0.04	-0.21	0.00	0.04	-
SNAP 0.24	-0.40	-0.01	0.10	-0.36	-0.07	-0.32	-0.06	0.13	-0.25 -
MATHARDSHIP_1 0.13	-0.71	-0.06	0.17	-0.27	-0.07	-0.16	-0.03	0.07	-0.15 -
MATHARDSHIP_2 0.14	-0.68	-0.06	0.15	-0.31	-0.06	-0.14	-0.03	0.04	-0.16 -
MATHARDSHIP_3 0.20	-0.49	-0.06	0.26	-0.32	-0.03	-0.13	-0.07	-0.04	-0.06 -
MATHARDSHIP_4 0.03	-0.61	-0.06	0.24	-0.18	-0.05	-0.12	0.00	-0.01	-0.07 -
MATHARDSHIP_5 0.04	-0.56	-0.05	0.11	-0.18	-0.02	-0.10	-0.03	-0.01	-0.06 -
MATHARDSHIP_6 0.16	-0.47	-0.05	0.22	-0.39	-0.02	-0.08	-0.15	0.05	-0.07 -
COLLECT 0.13	-0.50	-0.10	0.10	-0.22	-0.09	-0.05	-0.05	0.07	-0.13 -
REJECTED_1 0.23	-0.56	-0.02	0.11	-0.20	-0.04	0.01	0.00	0.09	-0.02 -
REJECTED_2 0.17	-0.66	-0.09	0.14	-0.11	-0.10	-0.08	-0.04	0.07	-0.07 -
ABSORBSHOCK	0.54	0.11	-0.09	0.07	0.05	0.07	-0.02	-0.05	0.03 0.12

SELFCONTROL_2	0.14	0.37	-0.08	0.01	0.07	0.00	0.14	0.02	0.01	0.02
SELFCONTROL_3	0.30	0.41	0.08	0.07	0.13	0.08	0.24	-0.02	0.10	0.03
OUTLOOK_1	0.14	0.14	-0.06	-0.01	0.08	0.06	0.04	0.00	0.02	-
0.02										
OUTLOOK_2	0.23	0.16	0.31	0.00	0.13	0.14	0.08	0.01	0.09	0.00
PEM	0.20	0.14	0.00	-0.06	0.09	-0.01	0.06	0.06	-	
0.18 0.06										
HOUSESAT	0.39	0.14	-0.28	0.13	0.13	0.05	0.02	0.00	0.01	0.23
SOCSEC1	0.21	0.01	-0.89	0.08	-0.05	-0.16	0.01	-0.22	-	
0.05 0.04										
SOCSEC2	0.21	0.02	-0.85	0.10	-0.03	-0.12	0.00	-0.24	0.02	0.05
SOCSEC3	-0.13	-0.02	0.82	0.05	0.05	0.10	-0.01	0.09	0.05	0.04
LIFEEXPECT	0.08	0.07	0.17	0.07	0.06	0.06	0.06	-0.01	0.15	0.03
HHEDUC	0.21	0.08	0.05	0.33	0.13	0.13	-0.02	0.02	0.77	0.09
KIDS_NoChildren	0.07	-0.01	-0.13	0.03	0.05	-0.06	-0.02	-0.87	-0.02	-
0.07										
KIDS_1	-0.11	0.06	0.43	-0.03	0.03	0.02	0.02	0.62	0.06	0.04
KIDS_2	-0.12	-0.02	0.36	-0.02	-0.08	0.05	0.04	0.75	-	
0.01 0.07										
KIDS_3	-0.10	-0.02	0.27	0.01	-0.07	0.07	-0.04	0.61	-	
0.05 0.13										
KIDS_4	-0.04	0.01	-0.09	0.06	-0.03	0.11	0.02	0.35	0.03	0.05
EMPLOY	0.03	0.01	-0.66	-0.12	-0.05	-0.34	-0.04	-0.01	-0.11	-
0.08										
RETIRE	0.27	0.04	-0.81	0.05	0.00	-0.12	0.01	-0.19	-	
0.03 0.03										
MILITARY	0.13	0.05	-0.35	0.09	-0.06	0.01	0.00	-0.02	0.00	0.03
Military_Status	-0.12	-0.06	0.33	-0.06	0.07	0.00	0.01	0.02	-0.01	-
0.01										
agecat	0.21	0.00	-0.87	0.14	-0.10	-0.04	-0.01	-0.09	-	
0.07 0.12										
generation	-0.19	0.00	0.83	-0.15	0.11	-0.03	0.00	0.08	0.09	-
0.14										
PPEDUC	0.19	0.11	0.05	0.32	0.13	0.17	0.00	0.01	0.81	0.05
PPETHM	-0.08	-0.08	0.19	-0.18	-0.05	-0.10	0.10	0.02	-0.09	-
0.02										
PPHHSIZE	-0.15	-0.03	0.40	-0.04	0.04	-0.05	-0.01	0.61	-0.04	-
0.01										
PPINCIMP	0.40	0.10	0.13	0.31	0.07	0.32	-0.02	0.00	0.39	0.25
PPMARIT	-0.21	-0.03	0.46	-0.17	0.08	-0.18	-0.07	-0.45	-0.01	-
0.31										
PPREG4	0.02	0.02	0.01	0.00	-0.03	-0.04	0.03	0.02	0.03	0.05
PPREG9	0.01	0.02	0.01	0.00	-0.02	-0.03	0.03	0.02	0.00	0.03
PPT18OV	-0.12	-0.05	0.23	-0.04	0.09	-0.09	-0.02	0.06	-0.08	-
0.11										
PCTLT200FPL	-0.08	0.00	-0.02	-0.12	-0.06	-0.11	0.03	0.03	-0.09	-
0.09										
finalwt	-0.13	-0.09	0.52	-0.18	0.00	-0.03	0.01	0.01	-	

0.43	0.05										
		RC10	RC12	RC11	RC26	RC20	RC8	RC14	RC6	RC15	RC22
sample		-0.03	0.07	-0.01	0.05	0.23	0.11	0.03	0.08	-	
0.02	0.00										
fpl		0.01	0.04	-0.01	-0.01	-0.18	-0.04	-0.10	-0.03	0.01	0.10
SWB_1		0.04	0.66	-0.07	0.03	0.06	-0.01	0.00	-0.01	0.06	-
0.06											
SWB_2		0.03	0.76	0.02	0.02	0.07	0.03	0.02	0.00	0.06	-
0.03											
SWB_3		0.03	0.70	0.09	-0.03	0.09	-0.02	-0.06	-0.06	0.05	0.01
FWBscore		-0.01	0.09	-0.03	-0.02	-0.01	0.00	-0.04	0.01	-0.05	-
0.02											
FWB1_1		-0.04	0.11	0.05	0.14	0.00	0.05	-0.04	0.04	0.06	-
0.02											
FWB1_2		0.00	0.24	0.09	0.15	-0.06	0.01	-0.03	0.03	0.08	-
0.02											
FWB1_3		-0.02	-0.20	0.06	-0.06	-0.01	0.01	0.02	-0.01	0.09	0.06
FWB1_4		-0.01	0.22	0.06	0.14	0.00	0.03	-0.02	0.02	0.08	-
0.04											
FWB1_5		0.00	0.00	0.01	-0.18	-0.01	0.01	-0.01	0.00	0.05	-
0.01											
FWB1_6		-0.03	-0.13	0.00	-0.11	0.00	-0.02	0.00	-0.03	0.17	0.09
FWB2_1		0.02	-0.04	0.02	0.07	0.05	0.01	0.02	0.01	0.06	-
0.03											
FWB2_2		0.00	0.03	0.03	0.01	-0.04	0.02	-0.08	0.03	0.08	-
0.01											
FWB2_3		0.04	0.04	0.06	0.17	0.04	0.00	0.10	0.03	-0.01	-
0.01											
FWB2_4		-0.02	-0.07	0.09	0.04	-0.02	-0.01	0.01	-0.02	0.10	0.02
FSscore		-0.01	0.05	-0.01	-0.02	-0.02	0.00	-0.02	-0.01	0.06	0.00
FS1_1		0.00	0.02	-0.03	-0.01	-0.02	0.00	0.01	-0.07	0.06	0.00
FS1_2		0.03	0.09	-0.01	-0.05	-0.13	-0.01	0.00	0.07	0.16	-
0.01											
FS1_3		-0.05	0.06	0.03	0.04	0.01	0.02	-0.01	0.08	-	
0.20	0.00										
FS1_4		-0.02	0.06	0.04	0.01	-0.01	0.02	-0.01	0.03	-0.09	-
0.03											
FS1_5		0.01	0.11	0.07	0.09	0.01	0.01	0.01	0.09	-	
0.13	0.00										
FS1_6		-0.02	-0.04	-0.10	-0.03	0.01	0.00	-0.02	-0.12	0.13	0.02
FS1_7		0.00	-0.02	-0.02	0.05	0.00	0.00	-0.01	-0.08	0.12	0.00
FS2_1		-0.02	0.12	-0.06	-0.18	-0.03	-0.01	-0.02	-0.01	0.43	0.03
FS2_2		0.00	0.11	-0.03	-0.12	-0.03	-0.02	0.02	0.00	0.55	0.02
FS2_3		0.02	-0.02	0.06	0.08	0.07	0.01	0.05	-0.10	0.59	0.01
SUBKNOWL1		-0.08	0.14	0.03	-0.05	-0.03	0.00	-0.01	0.17	-	
0.26	0.01										
ACT1_1		-0.04	0.05	-0.06	-0.28	-0.11	0.00	-0.05	-0.05	0.19	0.02
ACT1_2		0.01	0.08	0.01	-0.01	-0.04	-0.01	-0.03	-0.01	0.06	-

0.02										
FINGOALS	0.02	0.09	0.01	-0.09	-0.10	0.06	0.02	0.09	0.00	0.03
PROPPLAN_1	0.01	0.00	0.00	-0.07	0.09	0.03	0.00	-0.03	0.08	-
0.01										
PROPPLAN_2	-0.03	0.04	-0.01	0.01	0.08	0.01	0.00	0.04	0.03	-
0.01										
PROPPLAN_3	0.01	0.09	0.04	0.02	-0.01	0.03	0.03	0.00	0.02	-
0.02										
PROPPLAN_4	-0.01	0.09	0.03	0.10	0.06	0.00	0.00	0.06	-	
0.04 0.02										
MANAGE1_1	-0.10	-0.04	-0.09	-0.29	0.05	-0.02	-0.11	-0.06	0.09	0.01
MANAGE1_2	-0.01	-0.02	-0.04	-0.05	0.12	-0.03	-0.04	0.01	0.07	-
0.02										
MANAGE1_3	-0.05	-0.04	0.01	0.09	0.04	-0.03	-0.01	-0.02	0.10	-
0.02										
MANAGE1_4	-0.11	-0.01	-0.13	-0.17	0.03	-0.06	-0.09	0.00	0.04	0.05
SAVEHABIT	0.00	0.01	-0.01	0.07	0.07	0.00	-0.01	-0.02	0.12	0.00
FRUGALITY	-0.04	0.04	-0.28	-0.17	0.04	-0.02	-0.05	-0.07	0.20	0.00
AUTOMATED_1	0.03	-0.02	-0.04	-0.04	-0.01	0.01	0.00	-0.04	0.01	-
0.01										
AUTOMATED_2	0.01	0.01	0.00	0.02	0.01	0.00	0.01	-0.02	-	
0.02 0.00										
ASK1_1	-0.10	0.08	-0.04	-0.09	0.06	0.03	-0.02	0.12	-	
0.09 0.01										
ASK1_2	0.05	0.10	0.09	0.10	-0.04	-0.01	0.01	0.15	0.48	-
0.01										
SUBNUMERACY2	0.04	0.03	0.03	0.04	-0.04	0.04	0.06	-0.01	-	
0.06 0.05										
SUBNUMERACY1	-0.01	0.08	0.01	-0.03	0.02	0.00	0.00	0.09	-	
0.22 0.01										
CHANGEABLE	-0.01	0.05	0.15	0.01	0.27	-0.03	-0.04	0.01	-0.09	-
0.06										
GOALCONF	-0.02	0.12	0.00	-0.03	-0.11	0.00	-0.03	0.02	-0.04	-
0.06										
LMscore	-0.03	-0.04	-0.03	-0.01	0.19	-0.06	-0.11	-0.10	0.02	-
0.09										
FINKNOWL1	0.00	-0.02	0.04	0.06	-0.12	0.00	0.10	0.13	-	
0.05 0.16										
FINKNOWL2	-0.05	-0.04	-0.02	-0.01	0.11	-0.04	-0.04	-0.06	0.01	0.02
FINKNOWL3	-0.03	-0.06	-0.04	-0.03	0.15	-0.07	-0.11	-0.10	-0.02	-
0.03										
KHscore	-0.04	0.00	-0.01	-0.02	-0.18	0.02	0.01	0.04	0.00	0.05
KHKNOWL1	-0.15	0.08	0.00	0.03	-0.04	0.00	0.05	0.04	-	
0.02 0.21										
KHKNOWL2	-0.01	0.00	-0.10	-0.25	-0.12	0.02	-0.05	-0.07	0.04	-
0.01										
KHKNOWL3	0.04	-0.02	-0.03	0.06	-0.09	0.00	-0.04	-0.01	0.02	0.02
KHKNOWL4	-0.03	0.03	0.01	0.01	0.39	0.05	0.01	-0.06	-	

0.06 0.03										
KHKNOWL5	0.03	-0.01	0.02	0.10	0.29	0.00	-0.04	-0.16	-0.02	-
0.03										
KHKNOWL7	0.01	-0.05	0.00	0.04	-0.04	0.04	0.06	-0.09	-	
0.03 0.01										
KHKNOWL8	0.03	-0.07	0.02	-0.26	0.01	-0.02	0.02	0.10	0.06	-
0.11										
KHKNOWL9	-0.02	-0.01	-0.01	0.09	0.27	-0.05	0.02	-0.04	-	
0.03 0.05										
ENDSMEET	0.04	-0.03	0.01	0.14	0.06	0.01	0.08	0.04	0.02	0.02
HOUSING	0.19	-0.02	0.09	-0.06	0.16	0.00	0.01	0.04	0.02	-
0.04										
LIVINGARRANGEMENT	0.77	0.03	0.01	-0.02	0.05	0.00	0.00	-0.04	-	
0.05 0.11										
HOUSERANGES	0.07	-0.01	0.09	-0.08	0.14	0.00	-0.04	0.12	0.01	-
0.03										
VALUERANGES	-0.02	0.00	-0.01	0.04	-0.07	0.11	0.03	-0.09	-	
0.01 0.09										
MORTGAGE	-0.04	0.01	-0.01	-0.05	-0.03	0.03	0.00	0.02	0.00	-
0.02										
SAVINGSRANGES	-0.01	-0.06	-0.02	-0.09	0.10	0.00	0.01	-0.03	0.04	0.05
CONSPROTECT1	-0.03	-0.04	-0.02	0.37	-0.01	0.02	-0.04	0.31	0.09	0.00
CONSPROTECT2	-0.01	0.03	0.00	0.01	-0.03	-0.01	-0.03	0.73	-	
0.04 0.01										
CONSPROTECT3	0.00	-0.09	0.08	0.18	0.05	0.01	-0.04	0.74	0.05	0.03
EARNERS	0.74	0.05	0.00	-0.02	-0.07	-0.02	-0.03	-0.05	-	
0.09 0.06										
VOLATILITY	0.14	0.04	-0.01	0.02	-0.03	0.00	0.17	0.04	-	
0.05 0.07										
SNAP	0.03	-0.08	0.04	0.15	0.04	0.00	0.17	0.05	-0.03	-
0.12										
MATHARDSHIP_1	-0.02	-0.08	0.06	0.35	0.05	0.02	0.02	0.09	0.02	-
0.05										
MATHARDSHIP_2	0.00	-0.09	0.06	0.38	-0.01	0.00	0.01	0.08	0.01	-
0.02										
MATHARDSHIP_3	0.12	-0.07	0.12	0.49	0.08	0.00	-0.04	0.17	0.02	0.01
MATHARDSHIP_4	0.04	-0.07	0.07	0.44	0.02	0.03	-0.03	0.11	0.05	0.04
MATHARDSHIP_5	0.06	-0.06	0.04	0.51	0.01	-0.02	-0.05	0.14	0.05	0.02
MATHARDSHIP_6	0.05	0.01	0.16	0.55	-0.01	0.02	-0.02	0.19	0.00	0.00
COLLECT	-0.02	0.01	0.05	0.18	0.01	0.03	0.04	0.09	-0.08	-
0.02										
REJECTED_1	0.00	0.02	0.08	0.15	-0.09	-0.01	-0.06	0.09	-	
0.07 0.01										
REJECTED_2	0.00	-0.03	0.03	0.14	-0.08	0.02	-0.06	0.02	-0.09	-
0.01										
ABSORBSHOCK	-0.06	0.05	0.02	-0.01	-0.05	-0.01	0.01	0.01	0.01	0.04
BENEFITS_1	0.01	0.02	0.02	-0.09	-0.03	0.00	0.01	0.03	0.01	0.02
BENEFITS_2	-0.02	0.00	-0.01	-0.03	-0.08	-0.03	0.06	0.01	0.01	0.02

BENEFITS_3	0.00	0.00	0.01	-0.04	-0.02	0.02	-0.10	0.14	0.01	0.01
BENEFITS_4	0.01	0.04	0.05	0.11	-0.03	-0.07	-0.03	0.09	-0.01	-
0.03										
BENEFITS_5	-0.05	0.04	-0.02	-0.08	-0.06	-0.04	0.05	0.02	-	
0.01 0.04										
FRAUD2	-0.02	-0.01	0.02	0.10	0.01	0.04	-0.03	0.04	0.02	-
0.01										
COVERCOSTS	0.04	0.03	0.05	0.01	0.04	-0.04	0.00	0.04	-	
0.02 0.05										
BORROW_1	0.06	0.03	0.11	-0.19	0.02	0.03	0.03	0.09	0.11	-
0.08										
BORROW_2	0.03	0.01	0.04	-0.10	-0.03	0.04	0.04	0.04	0.12	-
0.10										
MANAGE2	-0.72	0.01	0.02	-0.09	-0.04	0.00	0.01	0.00	-	
0.19 0.08										
PAIDHELP	0.58	-0.04	0.11	0.13	-0.03	0.08	0.10	0.22	0.11	-
0.11										
HSLOC	-0.02	0.10	-0.02	0.00	0.80	0.10	0.22	-0.02	0.03	0.12
PAREduc	-0.03	0.01	0.03	-0.02	-0.09	0.00	0.02	0.04	0.01	-
0.02										
FINSOC2_1	0.04	0.05	0.05	0.08	0.08	0.00	0.03	0.15	0.00	-
0.02										
FINSOC2_2	0.00	0.03	0.02	-0.03	0.05	-0.03	-0.01	-0.02	0.05	0.04
FINSOC2_3	0.01	0.08	0.07	0.02	-0.05	-0.01	-0.04	0.09	0.00	0.04
FINSOC2_4	0.03	0.04	0.00	-0.03	0.03	-0.04	0.06	-0.02	0.02	-
0.01										
FINSOC2_5	0.03	0.10	0.02	-0.11	0.04	0.00	0.00	0.01	0.04	-
0.01										
FINSOC2_6	0.00	-0.02	0.03	0.01	-0.17	0.06	0.06	-0.12	-0.02	-
0.04										
FINSOC2_7	0.05	-0.03	0.02	0.04	-0.19	-0.06	0.02	-0.09	0.00	0.01
OBJNUMERACY1	-0.01	-0.03	0.13	0.03	0.02	-0.01	0.04	-0.03	0.01	-
0.02										
MATERIALISM_1	0.02	0.03	0.84	0.02	-0.03	0.01	0.00	0.01	0.04	0.00
MATERIALISM_2	-0.02	0.00	0.79	-0.01	0.08	0.00	0.00	0.00	0.01	0.02
MATERIALISM_3	0.05	0.03	0.85	0.06	-0.04	-0.02	0.02	0.04	0.00	-
0.02										
CONNECT	-0.04	0.06	-0.02	-0.07	-0.03	-0.03	0.00	0.00	-0.01	-
0.01										
HEALTH	0.03	0.40	0.02	-0.10	-0.12	0.00	0.09	0.06	0.01	0.00
SCFHORIZON	-0.04	0.02	0.02	0.07	-0.15	0.02	0.10	-0.01	0.05	0.09
DISCOUNT	-0.07	-0.03	-0.06	0.01	-0.14	0.00	0.00	-0.07	0.06	0.09
MEMLOSS	0.07	-0.30	0.08	0.24	0.07	0.01	-0.16	0.07	0.14	-
0.07										
DISTRESS	-0.01	-0.30	0.08	0.05	0.03	-0.03	0.00	-0.07	0.12	0.01
SELFCONTROL_1	0.03	-0.02	0.17	0.15	-0.02	0.04	-0.05	0.06	0.16	-
0.05										
SELFCONTROL_2	-0.02	0.10	-0.06	0.04	0.03	0.01	0.01	-0.02	-	

0.07 0.02										
SELFCONTROL_3	0.02	0.23	0.01	0.04	-0.02	0.00	0.05	0.00	-	
0.04 0.01										
OUTLOOK_1	0.03	0.10	0.06	-0.04	-0.06	0.04	0.05	0.12	0.08	-
0.03										
OUTLOOK_2	0.01	0.32	0.09	-0.05	-0.04	-0.01	0.01	0.05	-	
0.04 0.04										
PEM	0.00	0.34	0.10	-0.13	0.14	0.00	-0.16	-0.06	-	
0.09 0.12										
HOUSESAT	0.03	0.27	-0.05	-0.05	-0.05	0.00	0.03	-0.04	-0.03	-
0.02										
SOCSEC1	-0.06	-0.02	-0.02	-0.10	-0.04	-0.01	-0.14	0.00	0.01	-
0.06										
SOCSEC2	-0.04	-0.01	-0.01	-0.10	-0.02	-0.01	-0.09	0.00	0.01	-
0.04										
SOCSEC3	0.00	-0.02	-0.03	0.01	-0.03	-0.02	0.05	-0.02	0.00	0.08
LIFEEXPECT	0.02	0.22	0.00	-0.12	-0.13	0.08	0.17	0.07	0.04	0.09
HHEDUC	0.11	0.03	0.01	-0.02	0.03	0.01	0.01	0.05	0.04	0.02
KIDS_NoChildren	0.06	-0.02	0.02	-0.01	-0.01	0.01	0.00	0.00	0.00	-
0.28										
KIDS_1	0.00	0.10	0.07	-0.03	-0.07	0.05	0.09	-0.03	0.01	-
0.34										
KIDS_2	-0.09	0.07	-0.02	0.00	0.00	0.03	0.01	-0.03	-0.02	-
0.12										
KIDS_3	-0.06	0.03	-0.07	0.03	0.18	-0.02	-0.01	0.03	-	
0.01 0.29										
KIDS_4	0.05	-0.04	-0.03	0.04	0.06	-0.03	0.01	0.01	0.03	0.78
EMPLOY	0.06	-0.11	-0.04	0.09	-0.04	0.02	-0.12	-0.03	0.07	-
0.08										
RETIRE	-0.04	0.02	0.01	-0.06	-0.07	-0.01	-0.04	-0.01	0.01	-
0.04										
MILITARY	0.00	0.01	-0.01	0.01	-0.13	0.07	-0.86	0.04	-0.01	-
0.01										
Military_Status	0.01	-0.02	0.02	-0.03	0.12	-0.08	0.87	-0.04	0.02	0.00
agecat	-0.17	-0.05	-0.09	-0.04	0.02	-0.03	-0.11	0.01	0.00	0.14
generation	0.18	0.06	0.12	0.03	-0.01	0.03	0.06	-0.02	0.01	-
0.23										
PPEDUC	-0.07	0.01	0.01	-0.01	0.01	0.03	0.01	0.09	-0.01	-
0.02										
PPETHM	0.12	0.08	0.04	0.03	0.64	0.34	0.08	0.09	0.00	0.02
PPHHSIZE	0.37	0.05	0.06	-0.06	0.08	0.06	0.06	0.04	-	
0.02 0.27										
PPINCIMP	0.17	0.03	0.04	0.00	-0.12	-0.02	-0.03	-0.05	0.04	0.22
PPMARIT	-0.06	-0.07	0.09	-0.02	-0.04	0.01	0.15	0.07	0.01	-
0.07										
PPREG4	0.01	0.00	0.00	0.00	0.12	0.97	-0.06	0.00	-	
0.01 0.00										
PPREG9	0.01	0.00	-0.01	0.01	0.08	0.97	-0.05	-0.01	0.00	-

0.01											
PPT180V	0.51	0.01	0.09	-0.08	0.12	0.06	0.03	0.04	-		
0.02 0.60											
PCTLT200FPL	0.01	0.04	0.03	0.02	0.09	0.28	0.02	0.05	-0.04	-	
0.01											
finalwt	0.10	-0.01	0.01	0.03	0.08	0.03	0.13	-0.06	0.10	0.10	
	RC17	RC23	RC19	RC29	RC24	RC21	RC18	RC28	RC30	RC25	
sample	0.00	0.04	0.86	0.10	0.05	-0.03	-0.11	-0.01	0.07	0.00	
fpl	-0.08	-0.13	0.01	-0.03	-0.07	0.05	0.02	0.07	0.06	-	
0.02											
SWB_1	0.00	0.04	-0.03	-0.04	0.04	0.01	0.00	0.06	-		
0.01 0.02											
SWB_2	-0.01	0.02	0.03	0.03	0.07	-0.06	-0.04	0.00	0.03	0.02	
SWB_3	0.00	0.04	0.00	0.05	0.13	0.03	-0.08	-0.04	-0.03	-	
0.02											
FWBscore	-0.02	-0.01	-0.01	0.01	0.04	-0.13	-0.03	0.06	0.03	0.02	
FWB1_1	-0.05	0.01	0.04	0.01	0.04	0.16	0.02	-0.03	-0.03	-	
0.01											
FWB1_2	-0.01	0.00	0.03	0.00	0.07	0.09	0.01	-0.05	0.01	-	
0.02											
FWB1_3	0.01	0.04	0.00	-0.05	-0.05	0.23	0.06	-0.07	-0.06	-	
0.04											
FWB1_4	0.00	0.04	-0.01	0.01	0.06	0.10	0.01	-0.02	-0.02	-	
0.01											
FWB1_5	0.02	0.01	0.05	-0.06	-0.04	0.24	0.03	-0.08	-		
0.02 0.05											
FWB1_6	-0.01	0.07	-0.04	0.05	-0.03	0.20	0.05	-0.02	-0.04	-	
0.06											
FWB2_1	0.03	0.04	-0.01	-0.04	-0.04	0.09	0.03	-0.03	-0.04	-	
0.01											
FWB2_2	0.00	0.06	0.01	0.02	0.02	0.06	-0.02	0.01	0.01	-	
0.01											
FWB2_3	0.01	-0.07	0.05	0.04	0.00	0.02	0.04	-0.08	0.06	0.00	
FWB2_4	0.02	0.03	-0.01	0.00	-0.02	0.17	0.01	-0.01	-0.07	-	
0.07											
FSscore	-0.01	0.07	0.01	-0.03	0.02	-0.05	0.01	-0.01	0.02	0.01	
FS1_1	0.01	0.17	-0.01	-0.02	0.02	-0.04	-0.03	0.01	0.03	-	
0.02											
FS1_2	-0.02	-0.11	0.03	-0.09	0.09	-0.02	0.05	-0.04	0.03	0.04	
FS1_3	-0.01	-0.03	0.01	-0.03	0.02	0.06	0.01	-0.04	0.01	0.01	
FS1_4	-0.01	-0.01	-0.02	-0.06	0.03	0.03	0.03	-0.05	0.01	0.01	
FS1_5	-0.03	-0.11	0.01	-0.06	0.01	0.06	0.03	-0.06	0.01	0.03	
FS1_6	0.02	0.35	-0.01	-0.02	-0.01	-0.01	-0.03	0.00	0.04	-	
0.02											
FS1_7	0.02	0.24	-0.01	-0.02	0.03	0.03	0.01	-0.04	0.01	-	
0.03											
FS2_1	0.00	0.04	-0.01	0.07	-0.01	-0.22	-0.06	0.14	0.01	0.07	
FS2_2	-0.01	-0.04	0.03	0.04	0.03	-0.20	-0.03	0.10	0.06	0.07	

FS2_3 0.01	0.02	0.10	-0.03	0.03	0.01	0.13	0.02	-0.03	0.06	-
SUBKNOWL1 0.04 0.02	-0.04	-0.08	0.01	0.00	-0.02	0.05	-0.04	0.02	-	
ACT1_1 0.03	-0.02	0.19	-0.07	0.11	0.04	-0.03	-0.09	0.11	-0.16	-
ACT1_2 0.03	0.01	0.21	-0.01	0.03	0.03	0.02	-0.08	0.02	-0.05	-
FINGOALS 0.06	-0.07	-0.07	0.03	0.12	0.01	0.01	-0.08	-0.03	0.01	-
PROPPLAN_1	0.04	0.07	0.01	-0.06	0.01	-0.05	-0.02	0.06	0.00	0.06
PROPPLAN_2	0.03	0.11	0.01	-0.05	0.00	-0.03	0.00	0.01	0.05	0.04
PROPPLAN_3 0.03	0.00	0.02	0.01	0.01	0.05	-0.02	-0.02	-0.04	0.01	-
PROPPLAN_4	0.01	0.01	0.03	-0.07	-0.01	0.04	0.04	-0.05	0.04	0.01
MANAGE1_1 0.10	-0.03	0.16	-0.04	-0.05	0.05	0.11	-0.06	0.17	-0.22	-
MANAGE1_2 0.08 0.00	-0.01	0.25	-0.02	-0.07	0.02	0.10	-0.03	0.07	-	
MANAGE1_3 0.08	-0.10	0.15	0.01	0.02	0.04	0.24	0.02	0.01	-0.23	-
MANAGE1_4 0.10	0.00	0.14	-0.04	0.02	0.08	0.08	-0.04	0.17	-0.13	-
SAVEHABIT 0.07	0.00	0.21	0.00	0.03	0.07	0.17	0.01	-0.02	-0.08	-
FRUGALITY 0.07	0.03	0.37	-0.05	0.06	0.10	0.05	-0.06	0.17	-0.09	-
AUTOMATED_1 0.01	0.89	-0.01	-0.01	0.00	0.00	-0.02	-0.02	0.03	0.01	-
AUTOMATED_2	0.89	0.00	0.00	0.02	0.02	0.01	0.02	0.00	0.01	0.03
ASK1_1 0.03	0.00	0.07	-0.02	0.10	0.01	-0.01	-0.03	0.05	-0.17	-
ASK1_2 0.13 0.00	-0.03	-0.13	0.06	-0.05	0.08	0.06	0.09	-0.05	-	
SUBNUMERACY2 0.06	0.03	0.06	0.06	0.08	-0.07	-0.05	0.00	0.13	-0.23	-
SUBNUMERACY1 0.07 0.00	-0.05	0.02	0.02	0.05	-0.01	0.10	-0.04	0.03	-	
CHANGEABLE	0.00	0.09	0.01	-0.03	-0.10	0.49	-0.04	0.01	0.06	0.08
GOALCONF 0.05	0.00	0.15	0.03	0.01	0.07	0.00	-0.02	0.01	0.03	-
LMscore 0.09	-0.03	-0.06	-0.05	-0.03	0.08	-0.13	0.15	-0.05	0.06	-
FINKNOWL1 0.06 0.08	0.03	0.11	-0.04	0.11	0.02	-0.04	0.17	0.09	-	
FINKNOWL2 0.02	0.01	0.02	-0.07	-0.01	0.02	-0.10	0.10	0.05	0.01	-
FINKNOWL3	-0.04	-0.02	-0.06	0.03	0.18	-0.24	0.40	-0.04	0.03	-

0.10										
KHscore	-0.04	0.04	0.00	0.01	-0.03	0.02	-0.04	0.00	-	
0.01 0.06										
KHKNOWL1	-0.06	0.01	0.01	-0.03	0.01	0.00	-0.06	-0.03	-0.08	-
0.02										
KHKNOWL2	-0.07	-0.03	-0.02	0.00	0.00	-0.07	0.15	0.20	0.02	0.02
KHKNOWL3	-0.02	0.02	-0.02	-0.03	0.04	-0.01	0.57	0.04	-	
0.02 0.11										
KHKNOWL4	0.06	-0.06	0.06	0.04	0.04	0.00	0.16	-0.09	0.08	-
0.05										
KHKNOWL5	0.01	-0.07	-0.06	-0.06	0.13	-0.12	0.29	0.06	-0.14	-
0.14										
KHKNOWL7	0.02	0.03	0.00	0.03	0.01	0.06	0.18	-0.01	-0.06	-
0.01										
KHKNOWL8	0.07	0.06	-0.08	0.24	-0.14	0.01	0.46	0.17	0.17	-
0.08										
KHKNOWL9	0.02	-0.13	-0.03	0.12	0.07	-0.02	0.25	-0.04	-	
0.01 0.00										
ENDSMEET	0.02	-0.02	0.01	0.06	-0.02	0.00	-0.02	0.00	0.04	-
0.01										
HOUSING	0.07	0.01	0.01	-0.02	-0.01	-0.08	0.01	-0.10	-	
0.02 0.15										
LIVINGARRANGEMENT	0.02	0.00	0.00	0.05	-0.02	-0.02	0.06	0.00	-	
0.02 0.06										
HOUSERANGES	0.03	0.06	0.03	-0.02	-0.03	-0.12	0.00	-0.13	-	
0.04 0.14										
VALUERANGES	-0.03	-0.05	0.07	0.04	-0.01	0.09	0.01	0.03	0.03	-
0.09										
MORTGAGE	0.00	0.01	0.00	-0.01	0.02	-0.01	0.02	0.01	0.02	-
0.03										
SAVINGSRANGES	-0.02	0.04	0.02	0.08	-0.10	0.15	0.09	-0.09	-	
0.08 0.07										
CONSPROTECT1	-0.05	-0.07	-0.01	0.19	-0.04	0.06	-0.03	0.13	0.08	-
0.12										
CONSPROTECT2	-0.01	-0.09	0.03	-0.08	0.13	-0.03	0.09	-0.03	0.03	-
0.03										
CONSPROTECT3	-0.06	0.07	-0.02	0.18	0.03	0.04	-0.10	-0.04	0.03	0.02
EARNERS	-0.03	-0.03	-0.01	0.08	0.08	0.01	0.07	0.08	0.00	-
0.02										
VOLATILITY	-0.05	-0.12	0.06	0.42	-0.01	0.17	0.07	0.12	0.03	0.00
SNAP	0.14	0.00	0.06	0.19	0.10	0.01	0.04	-0.12	0.05	0.05
MATHARDSHIP_1	0.07	0.01	0.03	0.14	-0.03	-0.02	-0.03	0.03	-0.04	-
0.01										
MATHARDSHIP_2	0.05	0.02	0.03	0.14	-0.02	-0.03	-0.03	0.01	-	
0.03 0.01										
MATHARDSHIP_3	-0.01	0.00	0.02	0.12	-0.08	0.08	0.04	-0.06	-	
0.04 0.09										
MATHARDSHIP_4	0.00	0.07	-0.02	0.18	-0.09	-0.01	0.01	0.10	-0.11	-

0.02											
MATHARDSHIP_5	-0.01	0.05	-0.03	0.21	-0.07	0.00	-0.01	0.05	-		
0.10 0.04											
MATHARDSHIP_6	-0.05	-0.03	0.03	0.15	-0.04	0.08	0.08	-0.11	-		
0.02 0.06											
COLLECT	0.05	-0.08	0.10	0.39	0.01	-0.11	0.04	-0.14	0.14	0.07	
REJECTED_1	-0.04	-0.04	0.07	0.24	0.12	-0.11	-0.13	0.02	0.18	-	
0.05											
REJECTED_2	0.03	-0.04	0.07	0.22	0.07	-0.16	-0.09	0.05	0.20	-	
0.07											
ABSORBSHOCK	-0.04	-0.03	0.05	0.18	-0.06	0.15	0.24	-0.18	-		
0.08 0.13											
BENEFITS_1	-0.03	0.02	0.00	0.00	0.02	0.00	0.01	0.03	0.01	0.00	
BENEFITS_2	-0.11	-0.02	-0.02	0.01	0.04	0.01	0.04	0.03	-0.02	-	
0.06											
BENEFITS_3	-0.04	0.02	0.05	0.03	-0.04	0.00	-0.02	0.00	-		
0.05 0.03											
BENEFITS_4	0.02	0.05	0.09	-0.02	0.10	-0.08	-0.02	-0.02	-		
0.05 0.04											
BENEFITS_5	-0.03	-0.01	0.02	-0.02	0.04	-0.01	0.02	0.01	0.01	-	
0.03											
FRAUD2	0.03	0.04	0.04	0.69	-0.06	-0.06	0.03	-0.16	-		
0.08 0.01											
COVERCOSTS	0.02	0.06	0.03	-0.02	0.01	0.04	-0.01	0.00	0.56	0.03	
BORROW_1	0.05	0.01	0.01	-0.01	0.08	-0.10	-0.09	-0.02	0.04	-	
0.57											
BORROW_2	0.05	0.03	0.00	0.01	0.13	-0.03	0.03	0.04	0.05	0.67	
MANAGE2	-0.06	0.03	0.03	0.11	0.09	0.03	0.06	0.06	-		
0.08 0.04											
PAIDHELP	-0.05	0.07	0.00	0.00	0.14	0.18	-0.09	-0.12	0.06	-	
0.13											
HSLOC	0.00	0.05	0.00	0.01	-0.09	0.10	-0.11	0.03	0.03	-	
0.07											
PAREDOC	-0.02	0.01	-0.02	0.00	0.03	-0.03	0.01	-0.02	0.03	0.07	
FINSOC2_1	0.03	-0.01	-0.02	0.00	-0.02	-0.01	-0.03	0.02	-		
0.06 0.04											
FINSOC2_2	-0.02	0.04	0.01	-0.02	0.01	-0.03	0.02	0.00	-0.02	-	
0.03											
FINSOC2_3	0.00	0.04	0.04	-0.06	-0.04	0.02	0.02	-0.01	-		
0.04 0.00											
FINSOC2_4	0.01	0.08	0.03	0.01	0.03	-0.03	-0.01	-0.01	0.01	-	
0.03											
FINSOC2_5	-0.02	0.04	0.04	0.01	0.06	-0.06	-0.03	0.04	0.03	-	
0.04											
FINSOC2_6	-0.03	-0.11	-0.08	0.05	0.08	0.07	-0.01	0.04	0.09	0.12	
FINSOC2_7	-0.02	0.00	-0.04	-0.02	0.08	0.10	0.02	-0.02	0.02	0.05	
OBJNUMERACY1	0.04	-0.04	0.05	-0.06	0.03	-0.02	0.06	0.70	0.00	0.03	
MATERIALISM_1	-0.02	-0.01	0.01	0.01	0.03	0.01	-0.03	0.02	-0.01	-	

0.03											
MATERIALISM_2	0.00	0.01	0.01	0.00	0.05	0.08	-0.02	0.12	0.06	-	
0.01											
MATERIALISM_3	-0.02	-0.08	-0.05	0.03	0.02	0.03	0.01	0.00	-		
0.01 0.00											
CONNECT	-0.03	0.17	-0.07	-0.15	-0.08	0.03	0.07	0.45	-		
0.01 0.05											
HEALTH	0.00	0.21	0.02	-0.16	-0.06	0.09	0.11	-0.11	0.13	-	
0.04											
SCFHORIZON	-0.02	0.00	0.02	0.00	0.02	0.06	0.18	0.07	-0.08	-	
0.14											
DISCOUNT	-0.07	0.10	0.00	-0.02	0.05	0.11	0.09	0.07	-0.20	-	
0.09											
MEMLOSS	-0.03	0.00	-0.06	0.32	0.23	0.17	-0.27	0.06	0.03	0.03	
DISTRESS	0.02	-0.06	-0.07	0.21	0.15	0.12	-0.08	0.12	-		
0.02 0.03											
SELFCONTROL_1	-0.01	-0.29	-0.04	0.00	0.16	0.40	-0.06	-0.03	0.05	0.00	
SELFCONTROL_2	-0.01	0.64	0.02	-0.03	0.06	-0.02	0.06	-0.03	0.12	0.08	
SELFCONTROL_3	-0.03	0.40	0.05	0.00	0.12	0.02	0.01	0.01	0.09	-	
0.04											
OUTLOOK_1	0.05	0.07	0.09	-0.05	0.74	-0.05	0.03	0.00	-		
0.04 0.06											
OUTLOOK_2	-0.05	0.01	0.05	-0.02	0.52	0.08	0.02	-0.01	0.08	0.02	
PEM	-0.02	-0.05	-0.04	0.01	0.23	0.24	-0.06	0.05	0.17	-	
0.02											
HOUSESAT	-0.01	0.03	-0.05	-0.08	0.21	0.06	-0.02	0.11	0.10	-	
0.06											
SOCSEC1	0.00	0.01	-0.05	-0.05	0.01	0.01	0.01	0.05	-		
0.05 0.02											
SOCSEC2	-0.01	0.01	-0.05	-0.04	0.02	0.02	0.02	0.03	-		
0.05 0.01											
SOCSEC3	0.00	-0.02	0.00	0.01	-0.02	-0.01	0.03	0.01	0.08	-	
0.02											
LIFEEXPECT	-0.03	0.11	0.13	-0.10	-0.06	0.00	0.15	0.10	0.34	-	
0.15											
HHEDUC	-0.01	0.03	0.03	0.03	0.03	-0.06	-0.03	0.02	0.00	0.04	
KIDS_NoChildren	0.03	0.02	-0.02	0.02	0.03	-0.02	0.00	0.02	0.02	0.07	
KIDS_1	0.01	0.01	0.02	0.00	0.00	0.03	0.01	0.03	-0.19	-	
0.04											
KIDS_2	0.00	-0.02	0.01	0.03	0.04	0.03	0.00	0.02	0.04	0.01	
KIDS_3	-0.06	0.03	0.01	0.03	0.01	-0.03	-0.05	-0.01	0.15	0.00	
KIDS_4	-0.02	0.00	0.00	0.01	-0.01	0.00	0.00	0.00	0.08	-	
0.10											
EMPLOY	0.05	0.02	0.04	-0.03	-0.10	-0.11	-0.02	-0.11	-		
0.16 0.06											
RETIRE	-0.03	-0.01	-0.04	-0.06	-0.01	0.02	0.03	-0.03	-		
0.06 0.00											
MILITARY	-0.01	0.01	-0.01	0.01	0.00	0.04	0.05	0.01	0.01	-	

0.02											
Military_Status	0.01	0.02	-0.01	0.02	0.06	-0.02	-0.02	0.06	0.03	0.00	
agecat	-0.03	0.01	-0.02	0.00	0.00	0.04	0.00	0.08	0.03	-	
0.01											
generation	0.05	-0.01	-0.02	-0.02	0.02	-0.04	0.00	-0.10	-		
0.08 0.06											
PPEDUC	-0.03	0.01	0.04	0.04	0.01	-0.06	-0.03	-0.03	-		
0.04 0.03											
PPETHM	-0.03	0.01	0.13	-0.02	-0.02	0.07	-0.10	0.03	-		
0.03 0.05											
PPHHSIZE	0.01	0.07	0.02	-0.02	0.02	-0.05	-0.02	-0.06	-		
0.09 0.09											
PPINCIMP	-0.06	-0.15	0.00	0.02	-0.05	0.03	-0.01	0.02	0.07	0.00	
PPMARIT	0.07	0.02	0.05	-0.04	0.03	-0.03	-0.02	-0.11	0.03	0.05	
PPREG4	0.00	0.00	-0.06	0.03	0.01	0.00	-0.01	-0.01	-		
0.01 0.02											
PPREG9	0.01	-0.01	-0.08	0.02	0.02	-0.01	-0.01	-0.01	-0.02	-	
0.01											
PPT180V	0.04	0.06	0.03	-0.02	0.00	-0.10	-0.02	-0.08	-		
0.05 0.11											
PCTLT200FPL	0.01	0.02	-0.86	0.00	-0.07	-0.02	-0.02	-0.04	0.00	0.02	
finalwt	0.01	-0.05	-0.08	-0.01	-0.02	-0.02	-0.01	0.00	0.10	0.00	
	h2	u2	com								
sample	0.96	0.037	1.7								
fpl	0.81	0.186	6.5								
SWB_1	0.75	0.250	2.5								
SWB_2	0.76	0.241	1.7								
SWB_3	0.67	0.330	1.8								
FWBscore	0.97	0.033	1.3								
FWB1_1	0.73	0.267	1.9								
FWB1_2	0.72	0.281	2.7								
FWB1_3	0.72	0.282	1.7								
FWB1_4	0.73	0.274	2.4								
FWB1_5	0.60	0.396	1.7								
FWB1_6	0.65	0.354	1.8								
FWB2_1	0.77	0.234	1.3								
FWB2_2	0.73	0.268	1.5								
FWB2_3	0.74	0.263	1.5								
FWB2_4	0.64	0.355	1.4								
FSscore	0.96	0.040	1.5								
FS1_1	0.79	0.207	1.7								
FS1_2	0.69	0.307	2.0								
FS1_3	0.79	0.214	1.3								
FS1_4	0.78	0.225	1.4								
FS1_5	0.75	0.254	1.5								
FS1_6	0.74	0.260	2.6								
FS1_7	0.76	0.241	2.5								
FS2_1	0.69	0.314	3.0								

FS2_2	0.74	0.256	2.8
FS2_3	0.68	0.316	2.9
SUBKNOWL1	0.70	0.302	2.2
ACT1_1	0.63	0.369	6.7
ACT1_2	0.74	0.259	2.8
FINGOALS	0.52	0.475	2.9
PROPPLAN_1	0.68	0.323	1.4
PROPPLAN_2	0.76	0.237	1.5
PROPPLAN_3	0.78	0.225	1.9
PROPPLAN_4	0.78	0.225	1.9
MANAGE1_1	0.77	0.227	5.5
MANAGE1_2	0.64	0.364	4.5
MANAGE1_3	0.67	0.325	3.1
MANAGE1_4	0.52	0.483	7.9
SAVEHABIT	0.64	0.363	3.8
FRUGALITY	0.51	0.487	7.2
AUTOMATED_1	0.84	0.159	1.1
AUTOMATED_2	0.83	0.171	1.1
ASK1_1	0.50	0.495	3.2
ASK1_2	0.47	0.528	3.8
SUBNUMERACY2	0.30	0.704	5.1
SUBNUMERACY1	0.58	0.424	3.8
CHANGEABLE	0.49	0.513	3.5
GOALCONF	0.67	0.335	3.1
LMscore	0.88	0.124	1.6
FINKNOWL1	0.52	0.483	2.8
FINKNOWL2	0.71	0.286	1.4
FINKNOWL3	0.71	0.291	3.6
KHscore	0.92	0.084	1.4
KHKNOWL1	0.49	0.507	2.5
KHKNOWL2	0.65	0.354	2.3
KHKNOWL3	0.39	0.612	1.4
KHKNOWL4	0.53	0.472	2.9
KHKNOWL5	0.56	0.439	6.1
KHKNOWL7	0.43	0.569	1.7
KHKNOWL8	0.52	0.479	4.8
KHKNOWL9	0.62	0.383	2.9
ENDSMEET	0.79	0.210	1.4
HOUSING	0.86	0.137	4.0
LIVINGARRANGEMENT	0.66	0.336	1.2
HOUSERANGES	0.59	0.411	1.8
VALUERANGES	0.80	0.197	2.7
MORTGAGE	0.86	0.136	1.2
SAVINGSRANGES	0.53	0.473	2.9
CONSPROTECT1	0.46	0.545	5.4
CONSPROTECT2	0.69	0.315	1.6
CONSPROTECT3	0.75	0.250	1.9
EARNERS	0.63	0.372	1.3

VOLATILITY	0.45	0.548	5.3
SNAP	0.72	0.281	8.1
MATHARDSHIP_1	0.85	0.147	2.7
MATHARDSHIP_2	0.85	0.149	2.9
MATHARDSHIP_3	0.85	0.154	5.2
MATHARDSHIP_4	0.78	0.222	3.3
MATHARDSHIP_5	0.73	0.268	3.2
MATHARDSHIP_6	0.91	0.090	5.0
COLLECT	0.64	0.356	4.7
REJECTED_1	0.62	0.376	3.6
REJECTED_2	0.72	0.282	2.6
ABSORBSHOCK	0.53	0.472	3.1
BENEFITS_1	0.82	0.176	1.3
BENEFITS_2	0.82	0.176	1.8
BENEFITS_3	0.64	0.364	2.5
BENEFITS_4	0.75	0.249	1.8
BENEFITS_5	0.90	0.099	2.1
FRAUD2	0.56	0.437	1.4
COVERCOSTS	0.42	0.583	1.7
BORROW_1	0.54	0.464	2.7
BORROW_2	0.59	0.406	1.7
MANAGE2	0.67	0.334	1.6
PAIDHELP	0.75	0.250	4.5
HSLOC	0.82	0.183	1.6
PAREduc	0.67	0.335	1.7
FINSOC2_1	0.67	0.327	1.4
FINSOC2_2	0.82	0.183	1.2
FINSOC2_3	0.74	0.260	1.2
FINSOC2_4	0.76	0.242	1.1
FINSOC2_5	0.76	0.244	1.4
FINSOC2_6	0.48	0.524	2.4
FINSOC2_7	0.62	0.385	2.7
OBJNUMERACY1	0.54	0.463	1.2
MATERIALISM_1	0.74	0.257	1.1
MATERIALISM_2	0.67	0.330	1.2
MATERIALISM_3	0.77	0.226	1.1
CONNECT	0.48	0.524	4.4
HEALTH	0.53	0.466	7.3
SCFHORIZON	0.33	0.666	4.9
DISCOUNT	0.39	0.608	5.8
MEMLOSS	0.62	0.385	10.3
DISTRESS	0.52	0.476	4.8
SELFCONTROL_1	0.46	0.536	5.6
SELFCONTROL_2	0.66	0.345	2.2
SELFCONTROL_3	0.61	0.393	5.3
OUTLOOK_1	0.67	0.326	1.5
OUTLOOK_2	0.64	0.361	4.1
PEM	0.48	0.525	9.2

HOUSESAT	0.51	0.491	6.4
SOCSEC1	0.97	0.026	1.5
SOCSEC2	0.89	0.112	1.5
SOCSEC3	0.74	0.261	1.2
LIFEEXPECT	0.43	0.575	8.7
HHEDUC	0.82	0.179	1.9
KIDS_NoChildren	0.88	0.118	1.3
KIDS_1	0.77	0.229	3.1
KIDS_2	0.75	0.246	1.7
KIDS_3	0.65	0.352	2.7
KIDS_4	0.79	0.212	1.6
EMPLOY	0.71	0.288	2.5
RETIRE	0.81	0.190	1.5
MILITARY	0.93	0.066	1.5
Military_Status	0.93	0.068	1.5
agecat	0.93	0.066	1.5
generation	0.93	0.072	1.8
PPEDUC	0.88	0.124	1.8
PPETHM	0.72	0.281	2.7
PPHHSIZE	0.82	0.180	3.6
PPINCIMP	0.73	0.269	7.2
PPMARIT	0.71	0.290	4.9
PPREG4	0.97	0.032	1.1
PPREG9	0.96	0.041	1.0
PPT18OV	0.80	0.199	3.2
PCTLT200FPL	0.90	0.101	1.5
finalwt	0.60	0.403	3.2

	RC1	RC4	RC2	RC3	RC7	RC27	RC9	RC5	RC16	RC13	RC10
SS loadings	16.72	9.87	8.42	7.54	4.78	4.20	3.78	3.30	3.21	3.07	2.81
Proportion Var	0.11	0.07	0.06	0.05	0.03	0.03	0.03	0.02	0.02	0.02	0.02
Cumulative Var	0.11	0.18	0.24	0.29	0.32	0.35	0.37	0.40	0.42	0.44	0.46
Proportion Explained	0.16	0.10	0.08	0.07	0.05	0.04	0.04	0.03	0.03	0.03	0.03
Cumulative Proportion	0.16	0.26	0.34	0.41	0.46	0.50	0.54	0.57	0.60	0.63	0.66
	RC12	RC11	RC26	RC20	RC8	RC14	RC6	RC15	RC22	RC17	RC23
SS loadings	2.79	2.57	2.56	2.28	2.22	2.04	1.87	1.81	1.80	1.79	1.73
Proportion Var	0.02	0.02	0.02	0.02	0.01	0.01	0.01	0.01	0.01	0.01	0.01
Cumulative Var	0.48	0.49	0.51	0.53	0.54	0.56	0.57	0.58	0.59	0.60	0.62
Proportion Explained	0.03	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
Cumulative Proportion	0.68	0.71	0.73	0.76	0.78	0.80	0.82	0.83	0.85	0.87	0.89
	RC19	RC29	RC24	RC21	RC18	RC28	RC30	RC25			
SS loadings	1.71	1.70	1.49	1.49	1.47	1.38	1.30	1.24			
Proportion Var	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01			
Cumulative Var	0.63	0.64	0.65	0.66	0.67	0.68	0.69	0.70			
Proportion Explained	0.02	0.02	0.01	0.01	0.01	0.01	0.01	0.01			
Cumulative Proportion	0.90	0.92	0.93	0.95	0.96	0.98	0.99	1.00			

Mean item complexity = 2.9

Test of the hypothesis that 30 components are sufficient.

The root mean square of the residuals (RMSR) is 0.01

Fit based upon off diagonal values = 1

2.4 4.8 Artes e Barroso

Devido a não encontrar esses dados para download a análise não foi possível

2.5 Morettin e Singer, 2023

Num estudo planejado para avaliar o nível de poluição atmosférica por meio de medidas de elementos químicos depositados em cascas de árvores, obtiveram-se observações da concentração Al, Ba, Cu, Fe, Zn, P, Cl, Sr e Ca entre outros elementos em 193 unidades da espécie *Tipuana tipu* na cidade de São Paulo. Esses dados estão disponíveis no arquivo `arvores.xls`